Mohamed Tahiri

AI Engineer | LLM agents, RAG, evaluation

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AI engineer who ships LLM systems to production and measures them: agents, retrieval, evaluation, CI and infrastructure. Sole engineer behind an AI assistant used across an Air France-KLM group company.

Experience

AI Engineer | Air France-KLM group, Miami, FL — Mar 2026 – Present

- Designed and shipped, as sole engineer, an AI agent that answers aircraft maintenance and company questions with cited sources (Claude; 11 read-only tools incl. hybrid search, exact lookups, text-to-SQL; photo and voice; EN/FR/ES). Live since March 2026, now company-wide; HR made it the intranet's front door for policy questions.
- Indexed 1.3M+ chunks from maintenance history, technical manuals and the intranet (live sources refreshed hourly): hybrid dense + sparse search in Qdrant, Voyage AI embeddings, rank fusion and reranking.
- Built retrieval and answer evaluations (77 labeled queries; 33 real multilingual questions, LLM-as-judge) to choose models, benchmarking 5 embedders and 5 rerankers: reranking lifted nDCG@10 from 0.68 to 0.84, and a model upgrade cut p50 latency from 8.5 to 5.6 s and cost per question by 18% with no quality loss.
- Made it reliable and cheap to run: 2,600+ tests behind a 99% branch-coverage gate, security scans in CI, Terraform (111 resources) and Ansible, Sentry and CloudWatch alerts, prompt caching.
- Built a Claude Code plugin that turns a short Q&A into on-brand training videos and interactive trainings for non-technical teams: script, offline text-to-speech, Remotion rendering.

Software Engineer Intern, Backend & Data | Napta, Paris — Sep 2024 – Feb 2025

- Built Python/Airflow ETL pipelines from APIs, SFTP, databases and CSV files into PostgreSQL (millions of rows).
- Built a GitLab-to-Slack merge-request bot that cut code-review turnaround by 50%, from idea to team adoption.
- Improved internal Flask tools, CI/CD pipelines and the React/Next.js product, in a team of seven senior engineers.

Algorithms Instructor, volunteer | UT'Code, Compiègne — Feb 2023 – Feb 2024

- Taught algorithms and data structures for contests; placed 6th of 70 teams (210 contestants) in a coding contest.

Projects

Course discovery platform | web, iOS and Android — 2024 – Present

- Top committer (1,100+ commits, team of 15): NestJS backend on Supabase PostgreSQL, auth and permissions, transaction-safe Algolia search sync, and most of the Terraform/Terragrunt infrastructure with daily drift checks.

SDKs and integrations for a non-profit timetable platform | Python, TypeScript — 2023 – Present

- Wrote the OpenAPI 3.1 spec, checked nightly against the live API, and code generators for open-source, typed Python and TypeScript SDKs: 100% test coverage, about 3,300 downloads a month on PyPI and npm.
- Took over and re-architected the Alexa skill (JavaScript to TypeScript, AWS Lambda, DynamoDB, EventBridge; 7 locales); main author of the Home Assistant integration (118 installs; core submission in review).

Autonomous agent experiment | Claude Code, Python — 2026

- Set up a Claude Code agent that researches, builds, tests and publishes web-data tools, with a human approval gate on any spending: 7 tools live on the Apify Store, 117 tests, an MCP server listed in the official MCP Registry.

Education

Université de technologie de Compiègne (UTC), Sorbonne University Alliance | France — 2022 – 2026

Master's degree in Computer Science (Diplôme d'ingénieur), Information Systems track.
Coursework: machine learning, deep learning, distributed systems, databases, algorithms, statistics.

Preparatory classes in mathematics and physics (MPSI) | Lycée Charlemagne, Paris — 2021 – 2022

Skills

AI & LLM: Claude API, agents and tool use, RAG, hybrid search, reranking, embeddings, Qdrant, Voyage AI, evaluation (IR metrics, LLM-as-judge), prompt caching, MCP, Claude Code, PyTorch
Programming: Python, TypeScript, SQL, Go, Java
Backend & data: NestJS, Next.js, React, Chainlit, Flask, Airflow, PostgreSQL, DuckDB, Supabase, Algolia
Infrastructure: Docker, Terraform, Ansible, AWS, GCP, GitHub Actions, GitLab CI, Sentry, pytest, Playwright
Languages: French (native), English (fluent, TOEIC 985/990)